

Security Operations Center (SOC) Analyst Lab Report

Bob's Dry Cleaners Network Forensics Investigation

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Case Study: *L0ne Sh4rk's Revenge*

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Executive Summary

This report documents a comprehensive Security Operations Center (SOC) investigation involving a suspected brute-force intrusion against Bob's Dry Cleaners. The investigation focused on identifying indicators of compromise, determining the scope of unauthorized access, evaluating attacker activity, and assessing the risk of data exposure involving sensitive customer information.

Analysis of Linux authentication logs, Windows workstation logs, and Cisco ASA firewall logs confirmed that the attack was deliberate and successful. The attacker initially targeted the SSH service running on the DMZ server "baboon-srv" (10.30.30.20) through repeated authentication attempts against privileged accounts. After numerous failed login attempts against the "root" and "bob" accounts, the attacker successfully authenticated using the "bob" account before escalating privileges through sudo commands.

Following successful compromise, the attacker executed reconnaissance activity, installed additional tools including nmap, initiated network scanning, established remote connections to internal systems, and utilized FTP services for possible data exfiltration. Evidence strongly indicates that multiple systems were compromised and that sensitive customer information may have been exposed.

Incident Overview

Objective

The purpose of this investigation was to:

- 1. Determine whether the failed login attempts represented a deliberate cyberattack.
- 2. Identify the source and target systems involved.
- 3. Determine whether systems were compromised.
- 4. Evaluate the potential exposure of sensitive customer data.
- 5. Reconstruct the attack timeline using collected forensic evidence.

Network Environment

Bob’s Dry Cleaners network consisted of the following segments:

Network Segment	IP Range	Purpose
Internal Network	192.168.30.0/24	Internal systems and workstations
DMZ	10.30.30.0/24	Public-facing services
Internet Simulation	172.30.1.0/24	External attacker network

Primary Systems Investigated

Hostname	IP Address	Role
baboon-srv	10.30.30.20	SSH, DNS, and NTP server

dog-ws	192.168.30.101	Windows workstation
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ant-fw	Firewall	Cisco ASA firewall
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rsyslog server	192.168.30.30	Centralized log collection
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Tools and Methodology

Investigation Tools

The following tools and log sources were utilized during the investigation:

Log Analysis Platform

- Logalyze
- Linux grep command-line filtering

Log Sources

- auth.log
- firewall.log
- workstations.log

Investigative Techniques

- Timeline reconstruction
- SSH authentication analysis
- Brute-force attack detection
- Privilege escalation investigation
- Firewall traffic analysis
- Remote access verification
- Data exfiltration review

Investigation Findings

Phase 1 – Authentication Log Analysis

Initial Event Discovery

Security staff identified an abnormal spike in failed SSH authentication attempts beginning on April 26, 2011, at approximately 17:56:50.

Analysis of authentication logs within the 17:50–18:15 timeframe revealed significant activity spikes at:

Time	Number of Events
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17:55	2,006
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18:08	3,604
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The elevated event volume strongly suggested automated malicious activity rather than normal user behavior.

Brute-Force Authentication Attempts

Further filtering of the authentication logs revealed repeated SSH authentication failures targeting the baboon-srv system.

Key Findings

Indicator	Result
Primary Target	baboon-srv

Attack Type	SSH brute-force attack
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Remote Source IP	172.30.1.77
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Targeted Accounts	root and bob
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Authentication Failure Events	31
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The attack pattern demonstrated systematic password-guessing behavior consistent with automated brute-force activity.

Failed Login Attempts

Detailed grep analysis identified the following failed login counts:

Account	Failed Attempts
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root	41
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bob	29
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The final failed login attempt occurred at 18:04:05.

The “root” account was targeted first, indicating that the attacker initially attempted to obtain administrative-level access directly before shifting focus to the “bob” account.

Phase 2 – Successful Compromise and Privilege Escalation

Successful Authentication

Authentication logs confirmed that the attacker successfully authenticated to baboon-srv using the “bob” account at approximately 18:04:07.

Additional logs showed a second successful login at 18:04:33, with the session remaining active until 18:14:53.

This sequence of failed logins followed immediately by successful authentication strongly confirms that the brute-force attack succeeded.

Privilege Escalation Activity

Following successful authentication, the attacker executed multiple sudo commands to elevate privileges and gain administrative control of the system.

Commands Observed

```
sudo /usr/sbin/tcpdump -nni eth0
```

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sudo /usr/bin/apt-get update
```

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sudo /usr/bin/apt-get install nmap
```

Significance of Commands

Command	Purpose
tcpdump	Network packet capture and traffic monitoring
apt-get update	Repository and package updates

apt-get install nmap Installation of reconnaissance tool

The ability to execute sudo commands confirms that the attacker successfully obtained elevated privileges.

Phase 3 – Network Reconnaissance and Log Gap Analysis

Seven-Minute Authentication Gap

Investigators identified a seven-minute gap in authentication logging activity following the successful compromise.

This gap likely occurred while the attacker performed internal reconnaissance and scanning operations.

Possible Attacker Activities During the Gap

- Internal network reconnaissance
- Port scanning
- Credential harvesting
- Malware installation
- Data staging
- Log manipulation attempts
- Lateral movement preparation

The timing of the gap coincided with evidence showing installation and execution of nmap.

Firewall Analysis

Firewall analysis revealed significant malicious network activity originating from the compromised environment.

Firewall Findings

Observation	Result
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Total Firewall Events	4,404
baboon-srv Related Events	17
Peak Event Volume	3,536 events
Peak Activity Time	18:08

Investigators identified heavy traffic targeting multiple systems over Remote Desktop Protocol (RDP).

Ports Identified

Port	Service
53	DNS
3389	RDP
80	HTTP
443	HTTPS
21	FTP

The firewall logs showed 512 events involving port 3389, strongly suggesting active scanning or attempted remote compromise of Windows systems.

Phase 4 – Workstation Compromise and Lateral Movement

Remote Login Investigation

Analysis of workstation logs associated with the “dog-ws” workstation confirmed successful remote access.

Key Findings

Indicator	Result
Workstation	dog-ws
Event ID	528
Login Type	Remote login
Successful Login Time	19:11:01 – 19:11:08

The successful remote login demonstrated lateral movement from the initially compromised server into the internal network.

Command Execution and File Transfer Activity

Further analysis identified execution of:

/system32/cmd.exe

/system32/ftp.exe

Security Concerns

Executable	Risk
cmd.exe	Interactive command execution
ftp.exe	Potential data exfiltration

The use of ftp.exe strongly suggests that files may have been transferred outside the network.

Firewall logs further confirmed an outbound FTP transfer to external host 170.30.1.77 over port 21 at approximately 19:11:39.

This activity represents a strong indicator of possible data exfiltration.

Timeline of Events

Time	Event
17:56:50	Initial SSH brute-force activity begins
17:57–18:00	Repeated failed root authentication attempts
18:00–18:04	Failed authentication attempts against bob account
18:04:05	Final failed login attempt
18:04:07	Successful login using bob account

18:04:33	Additional successful login established
18:05–18:12	Network scanning and reconnaissance activity
18:08	Peak firewall activity detected
19:11:01	Successful remote workstation login
19:11:39	FTP transfer to external system detected
18:45 onward	Continued network activity observed

Risk Assessment

Evidence of Compromise

The investigation confirmed the following:

- Successful SSH compromise of baboon-srv
- Privilege escalation through sudo access
- Installation of reconnaissance tools
- Internal network scanning
- Remote workstation compromise
- Execution of FTP services
- Potential data exfiltration

Potential Impact

Because Bob's Dry Cleaners stored customer credit card information and personal data, the compromise represents a significant business risk.

Potentially Exposed Information

- Customer credit card numbers
- Personally identifiable information (PII)
- Executive customer records
- Internal network architecture information
- Authentication credentials

The confirmed FTP activity significantly increases the likelihood that sensitive information was transferred outside the organization.

SOC Analyst Assessment

The collected evidence conclusively demonstrates that the incident was a deliberate and coordinated cyberattack. The attacker employed brute-force authentication techniques against SSH services exposed in the DMZ before successfully compromising the “bob” account and escalating privileges through sudo access.

Once elevated privileges were obtained, the attacker conducted reconnaissance, installed additional tools, scanned internal systems, established remote access sessions, and initiated outbound FTP communications. These activities are consistent with advanced post-exploitation behavior and indicate a multi-stage intrusion involving persistence, reconnaissance, lateral movement, and potential exfiltration.

The compromise extended beyond the initial DMZ server and impacted internal systems, including a Windows workstation. The evidence strongly suggests that sensitive organizational and customer data may have been exposed. The investigation highlights the importance of centralized logging, network segmentation, brute-force detection, privileged account monitoring, and outbound traffic inspection.